25900011

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION MARCH 2025
SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCBCH100 - ESSENTIALS OF BIOCHEMISTRY : VITAMINS, HORMONES, ENZYMES ANDNEUROTRANSMITTERS
(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I)and Appreciation(Ap)
Students should attempt at least one question from each course outcome to enhance their overalloutcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each question carries 1 mark

1. Name the vitamin that play an important role in wound repair, protects body against stress, increases phagocytic action of leucocytes.

ANS: d). Vitamin C [K] / [CO1]

2. Which fat soluble vitamin has antioxidant function?

ANS: b). Vitamin E **[U]** / [CO1]

3. The neuropeptide that is linked to the regulation of blood pressure and water balance in the body.

ANS: a). Vasopressin [K] / [CO4]

4. Choose the enzyme which contains a protein part and a cation.

ANS: c) Carbonic anhydrase [U] / [CO5]

5. Which class of enzymes catalyzes oxidation-reduction reactions?

ANS a). Oxidoreductases [K] / [CO5]

[1x5 = 5]

Part B
Fill In The Blank Questions
Answer all questions
Each question carries 2 marks

6. A person with a bleeding disorder due to a deficiency of clotting factorsis likely deficient in _____.

ANS: Vitamin K [U] / [CO1]

7. _____ are non protein molecules that bind to enzymesand assist in atalyzing biochemical reactions.

ANS: Cofactors [K] / [CO1]

8. Catecholamines are ---------- and----------

ANS: Hormones and Neurotransmitters. [K] / [CO4]

9. The pancreas produces two hormones namely ------ & ---- whichregulates blood glucose .

ANS: Insulin and Glucagon [K] / [CO3]

10. The Michaelis - Menten equation is used to describe the relationshipbetween the .....and substrate concentration

ANS:Reaction rate (velocity) [K] / [CO5]

[2x5 = 10]

Part C
Short Answer Questions
Answer any 5 questions
Each question carries 3 marks

11.Explain the letter-based nomenclature system of vitamins withexamples.

ANS : Mention the classification of vitamins (3 marks)[K] / [CO1]

12. Discuss the conditions occur due to the deficiency of hypothalamus?

ANS: Mention any two deficiencies (1.5 marks each) [U] / [CO3]

13. Describe the role of G-protein-coupled receptors (GPCRs) in peptide hormone signaling.

ANS: Explain the role of G-protein-coupled receptors (GPCRs) in

Peptidehormone signalling.(3marks) [K] / [CO3]

14. Explain classification of neurotransmitter based on their function.

ANS:Classication based on structure (1.5 marks)

Classification based on function (1.5Marks) [U] / [CO4]

15. Why do enzymes exhibit geometrical specificity?

ANS: Explain Geometrical specificity [U] / [CO5]

16. Draw the double reciprocal plot and also write the LB equation.

ANS: For Graph : 2 Marks

Equation : 1 marks [A] / [CO5]

17. What is the significance of enzyme specificity in diagnostic andresearch tools?

ANS: Explain enzyme specificity (3 marks) [E] / [CO5]

[3x5 = 15]

Part D
Short Essay Questions
Answer any 4 questions
Each question carries 5 marks

18. Write about vitamin B9 deficiency in detail.

ANS: Any 5 deficiencies (One mark each) [U] / [CO1]

19. Describe the process of intracellular signaling involved in steroidhormone action.

ANS: Explain the intracellular processing (5 marks) [K] / [CO3]

20. Explain the concept of target glands and describe how they areinvolved in the endocrine system's regulation of body functions. Providean example of how a target gland responds to hormonal signals.

ANS: Explain the concept (5 Marks) [U] / [CO2]

21. What is Km? Evaluate the MichaelisMenton equation when [S] equalsKm.

ANS: Definition (1.5 Marks)

Equation(1.5 Marks)

Explanation of [S]=Km (2marks) [U] / [CO5]

22. Explain what is the relationship between enzyme concentration and rate of the reaction

ANS: Explanation (3marks)

Graph (2marks) [U] / [CO5]

23. Summarize the effect of pH and temperature using an illustrated curve

ANS: Explanation (3marks)

Graph (2marks) [U] / [CO5]

[5x4 = 20]